

Trident: Tactical Research, Inspection, and
Debris ENGagement Tool (UUV)

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A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF
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ABSTRACT IN ENGLISH

Thailand's marine ecosystems are increasingly threatened by seabed pollution, habitat degradation, and insufficient infrastructure inspection due to the high cost and inaccessibility of traditional underwater monitoring systems. This project presents TRIDENT, a modular, low-cost Unmanned Underwater Vehicle (UUV) developed to support marine research, debris removal, and infrastructure maintenance in coastal regions. TRIDENT integrates a vision-based system for marine debris detection, a robotic arm module for interaction tasks, and high-resolution imaging for biodiversity documentation. Built using moldcast epoxy resin and silicone for cost-effective waterproofing, the vehicle incorporates an efficient ballast tank system and pressure sensors for quasi-static depth control. Realtime data acquisition is enabled via LAN communication with a Raspberry Pi and Arduino-based control architecture running ROS. With a compact and energy-efficient design, TRIDENT enhances underwater exploration in confined environments while addressing 11 critical marine challenges. The system contributes to sustainable ocean monitoring by reducing operational costs, improving inspection accessibility, and enabling data-driven conservation efforts.

ACKNOWLEDGEMENT

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We also thank the School of Engineering, KMITL, for providing the knowledge and resources that enabled us to carry out this work. The skills gained from various courses

greatly contributed to our design and implementation. Lastly, we appreciate the support from our families, friends, and teammates for their encouragement and collaboration during this journey.

Nabhatara Tongsri

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LIST OF ABBREVIATIONS AND SYMBOLS

UUV	Unmanned Underwater Vehicle
ROV	Remotely Operated Vehicle
AUV	Autonomous Underwater Vehicle

CHAPTER 1

INTRODUCTION

1.1 SIGNIFICANCE AND BACKGROUND OF THE PROBLEM

1.1.1 Noise Pollution in UUV

Traditional UUV came packaged with noise pollution, whether it being from its motors, acoustic sensors, or even water disperse around its shape. This toughens the recording and tracking attempt for various marine species.

Big commercial submarines adopt another method providing more control, and almost no noise is created. A ballast tank, which is by all mean a better method for a lengthy period hovering, compared to other UUV. But this is hard to implement in a small-scale submersible.

1.1.2 Enhancing Marine Species Monitoring in Thailand

Monitoring pollution levels and marine biodiversity is crucial for managing ecosystems sustainably. As markers of the health of the environment, marine species' loss frequently indicates more serious problems with habitat degradation, temperature changes, or water quality.

However, traditional maritime monitoring techniques that are commonly used in Thailand, such as diver-based surveys or equipment installed on boats, are risky, time-consuming, and labor-intensive. These techniques are further constrained by depth limitations, strong currents, and low vision.

1.1.3 Ensuring Maintenance of Underwater Systems

Underwater systems, for example, pipelines, fish farms, and marine research equipment, are constantly exposed to harsh environmental conditions including corrosion, and damage from marine life. These systems require regular inspection, repair, and maintenance to remain functional and safe.

Traditional inspection methods rely on divers or expensive Remotely Operated Vehicles (ROVs), which are costly, time-consuming, and pose safety risks especially in deep or murky waters. As a result, many installations suffer from delayed maintenance and undetected failures, leading to higher repair costs or system downtime.

1.1.4 Traditional UUV is costly

Unmanned Underwater Vehicles are essential for tasks such as marine research, environmental monitoring, and infrastructure inspection. However, most traditional UUVs are expensive and inaccessible to small research labs, universities, and local conservation groups. For instance, Chasing gladius Mini (35,000 –60,000 Thai baht), BlueROV2 (180,000 Thai baht), and VideoRay Pro 5 (500,000 – 3,700,000 Thai baht). These systems also require additional costs for operation, deployment, and training. Due to these barriers, many underwater systems go uninspected, and marine research is limited—especially in developing regions like Thailand.

1.2 OBJECTIVES OF THE PROJECT

The primary objective of this project is to design and develop multipurpose Unmanned Underwater Vehicle (UUV) that supports various underwater applications in both research and maintenance. The system is aimed to provide affordable, modular, and adaptable platform that can operate effectively in canal and coastal environments of Thailand. **Key Objectives**

1. Develop a Modular Multi-Use UUV Platform:

Create a compact and cost-effective UUV structure that allows easy integration of various tools and sensors based on mission requirements. Design the system to support plug-and-play modules for quick reconfiguration.

2. Enable Underwater Debris Detection:

Integrate vision-based detection for identifying marine litter and seabed debris. Attach basic cleaning tools or manipulators for collecting or tagging debris for environmental maintenance.

3. Support Marine Biological Research (Ichthyology):

Equip the UUV with high-resolution cameras to monitor marine life habitats and behaviors. Enable data collection for research in ichthyology and marine biodiversity mapping.

4. Monitor and Inspect Underwater Infrastructure:

Facilitate routine inspection of underwater systems such as pipelines, fish cages, research sensors, and marine installations. Provide real-time visual data for structural assessment and potential failure detection.

5. Reducing Noise Pollution in UUV:

Reduce the noise pollution created by other modern UUV, by reverting to a traditional mean. A fully controllable ballast tank and maneuverable hovering.

1.3 SCOPE AND LIMITATIONS OF PROJECT

To minimize manufacturing costs while maintaining structural functionality, the connecting components of the Unmanned Underwater Vehicle (UUV) were fabricated by 3D-printing PETG components before applying a resin shell around. This method allows for rapid prototyping, ease of customization, and acceptable water resistance, particularly in shallow-to-moderate depth applications. Traditional 3D-printed materials such as PLA, while accessible, exhibit poor waterproofing and dimensional stability in underwater environments, making them unsuitable for critical structural components.

Despite its advantages, PETG with resin coat exhibits lower mechanical properties. Specifically, tensile strength, impact resistance, and fatigue tolerance when compared to industrial-grade polymers (e.g., polycarbonate, PEEK) or metallic materials. These limitations must be considered in operational contexts involving hydrostatic pressures, mechanical vibrations, or potential collision events. Finite Element Analysis (FEA) is recommended for evaluating stress concentrations within these components under load conditions.

Given the spatial constraints inherent in the UUV's compact form factor, the onboard battery modules are limited in both physical size and capacity, leading to a reduction in operational endurance. This imposes constraints on mission duration and necessitates efficient power management strategies, such as pulse-width modulation (PWM) for thruster control and low-power sleep modes for auxiliary systems. Trade-offs between vehicle autonomy and power density are actively evaluated.

For underwater communication, Bluetooth and other RF-based protocols are infeasible due to rapid signal attenuation in conductive media like seawater. To overcome this, the UUV employs a tethered Ethernet-based Local Area Network (LAN) for high-throughput, low-latency communication during deployment and testing. This configuration supports real-time control, telemetry acquisition, and bulk data transfer, which are critical for debugging, mission uploads, and sensor data retrieval.

The vehicle is equipped with two ballast tanks designed to modulate its net buoyancy, allowing for depth control without constant thrust input. By achieving near-neutral buoyancy and increasing vertical displacement, the UUV can maintain submerged states for extended periods. This feature is advantageous for the UUV's primary objective: Reducing Noise Pollution in UUV. Reduced propulsion noise and acoustic signal created by other UUV.

1.4 Methodology Used in the Project

TRIDENT is designed to have a highly compact form factor, as its structure contains air, which increases buoyancy and makes submersion more challenging due to density differences. Since one of the primary objectives of the project is to navigate confined spaces, minimizing the vehicle's size enhances its ability to operate efficiently in such environments.

1.4.1 Software Development

The robot primarily operates using ROS (Robot Operating System), which provides a flexible framework for handling navigation, sensor integration, and autonomous decision-making. For underwater mapping, the robot employs Simultaneous Localization and Mapping (SLAM), a technique that allows it to construct a map of the seabed while simultaneously determining its own position within that environment. To achieve this, the system utilizes laser distance sensors, which play a crucial role in detecting the contours and features of the seabed in real time. These sensors, specifically infrared (IR) distance sensors from Sharp, are designed to measure distances within a range of 20 cm to 150 cm. They operate based on triangulation and reflected infrared light intensity, where the emitted IR beam is reflected off surfaces and detected by an internal position-sensitive detector (PSD). This data is then processed to estimate distances accurately, enabling the robot to create a detailed and precise underwater map while navigating autonomously.

1.4.2 Communications

The TRIDENT robot utilizes LAN communication for underwater operations, as water can completely block the radio frequency signals required for other forms of communication. Within the robot, a Raspberry Pi serves as the primary control unit, running ROS (Robot Operating System) to manage and facilitate communication with an Arduino Nano. The Arduino Nano is responsible for controlling the sensors and motors via UART (Universal Asynchronous Receiver-Transmitter). UART is a serial communication protocol that enables data transfer between devices without the need for a shared clock signal. Data is transmitted one bit at a time, including a start bit, data bits, an optional parity bit, and a stop bit. This protocol is commonly used in embedded systems, microcontrollers, and computers, supporting full-duplex communication through transmit (TX) and receive (RX) lines.

1.5 Project Implementation Plan

1.5.1 Independent Site Selection and Mission Planning

Site selection will be conducted independently based on publicly available data, prior studies, and observed environmental needs. Emphasis will be placed on identifying locations suitable for underwater inspection, ecological observation, and experimental data collection.

1.5.2 Data acquisition and analysis

The UUV would deploy to conduct comprehensive seabed surveys. Generate detailed monitoring of the underwater terrain, accurately documenting the distribution of pollutants and the characteristics of marine habitats.

1.5.3 Targeted Debris Removal Operations

UUV would deploy for targeted debris removal missions. The operations will focus on extracting pollutants, particularly plastics and abandoned fishing gear, from identified hotspots, thereby mitigating their adverse environmental impact.

1.5.4 Marine Species Documentation and Monitoring

The UUV will be employed to capture video and image data of marine species within their natural habitats. Which crucial for biodiversity assessments, species identification, and behavioral studies, contributing to a deeper understanding of the local marine ecosystems.

1.5.5 Underwater Infrastructure Inspection

Routine inspections of underwater infrastructure, including pipelines, fish cages, and other installations. These inspections will facilitate the detection of structural damage, corrosion, and other potential issues, enabling timely maintenance and preventing costly failures.

1.6 Benefits Derived from the Research

1.6.1 Reduce Seabed Pollution

The UUV's advanced mapping and debris removal capabilities will directly contribute to cleaner and healthier marine environments. By accurately identifying and targeting pollution hotspots, particularly those contaminated with plastic and abandoned fishing gear, the project will mitigate the detrimental effects of these pollutants on marine life.

1.6.2 Data-Driven Conservation

The UUV will provide data that supports informed decision-making in marine conservation and management. This data will enable targeted interventions and enhance the effectiveness of conservation efforts.

1.6.3 Improved Underwater Infrastructure Maintenance

By deploying the UUV for routine inspections of underwater infrastructure, such as pipelines and fish cages, the project will reduce the reliance on expensive and potentially hazardous diver-based inspections. Increasing driver safety.

1.7 Capstone Design Details

1.7.1 Application of Knowledge and Skills from Previous Courses

Subjects that lie closely to the project include

01006723: General Physics 1 classical mechanics, including motion, forces, energy, and buoyancy

01006725: General Physics 2 electronic circuits, understanding motor behavior, battery systems

01416312: Electric Circuit and Electronics designing and building the UUV's electronic systems, including power distribution, sensor connections, motor controllers

01416305: Artificial Intelligence Technology vision-based object detection

01416308: Kinematics and Dynamics analyzing forces and torques on the body, predicting behavior under water currents, and developing motion control algorithms

01416500: Computer Vision image processing, object detection, and navigation using onboard cameras

01416306: Safety and Standardization in RAI compliance with underwater safety standards (IP ratings, battery safety)

01416319: Manufacturing Process Applied in the mold-based component fabrication using epoxy resin and silicone, and in selecting suitable materials and production methods for waterproof, strong, and modular design

01416511: Mobile Robots autonomous navigation, localization, path planning, sensor fusion, and the integration of mechanical and software systems in mobile robotic platforms

01416408: COM AID Mechanical Design CAD modeling, stress analysis, and mechanical design optimization for components like the ballast tank, motor housing, and main frame structure

1.7.2 Design of a Device

For a modular use UUV platform, the robot must be compatible with a tool such as robot arm for collecting debris or holding on to pipeline. For inspection and monitoring, the device is equipped with a camera module, two hydrostatic pressure sensors and two laser distant sensors. These sensors enable the use of Point Cloud for mapping the underwater bed for more data collecting.

Additionally, focusing primarily on minimizing environmental impact, the design of our robot is centered around this concept. Two key improvements over traditional UUVs are (1) the elimination of noise pollution generated by the motor and (2) reducing the cost of the robot by optimizing its design. Additionally, the robot is designed to have a smooth, rounded shape with as few openings as possible to better withstand the challenges of Thailand's polluted waterways.

CHAPTER 2

RELATED THEORIES AND LITERATURE REVIEW

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- 2.2 LITERATURE REVIEW
- 2.3 SUMMARY

CHAPTER 3

MECHANICAL DESIGN

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CHAPTER 5

CONCLUSION AND FUTURE WORK

- 5.1 CONCLUSION
- 5.2 CONTRIBUTIONS
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Appendix A
ADDITIONAL MATERIALS

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